STEP 1 — Diagnose the complexity profile

Run five quantitative tests on the trained model + data, cheapest first. Each test flags one dimension; all cutoffs are provisional (Table 2).

Dimension	Diagnostic	Flag if
Collinearity	VIF, condition number κ(X)	VIF > 5 (moderate), > 10 (severe)
Interactions	SHAP interaction matrix	$\lvert\Phi_{jk}\rvert > 0.05 \times$ mean main effect
Nonlinearity	degree-2 polynomial vs. linear, ΔR^2	NL > 0.10
Suppressors	r vs. partial correlation ρ	sign flip, or $\lvert\rho\rvert > 2\lvert r\rvert$
Capacity	effective Hessian rank	leaves > 10^5 or rank > 100

Any test flagged?

NO

Complexity-simple

Standard SHAP, LIME, and IG are likely reliable.

YES

STEP 2 — Map diagnosis to method

If several tests flag, provisional tie-breaker (not a severity ranking):
interactions > suppressors > collinearity > nonlinearity > capacity

Interactions	TreeSHAP interaction values, Shapley–Taylor, Integrated Hessians
Suppressors	PatternAttribution / Haufe patterns; report the suppressor check
Collinearity	grouped interv. SHAP (model); obs. SHAP, CALIME, AEC (data)
Nonlinearity	IG, Integrated Hessians, DeepLIFT
Capacity	SmoothGrad, VarGrad, ensemble attributions
Architecture	SLALOM (transformers); applicability: failure-mode map (§3)

STEP 3 — Validate the diagnosis

1. Apply one recommended and one discouraged method — rankings should diverge on flagged features.
2. Model questions: ROAR — recommended method should give a steeper drop (confounded by redundancy).
3. Stability: perturb input by ε; recommended method's attributions should change less.
4. Agreement ($\tau > 0.9$) despite a flag → false positive or no effect (interactions: compare ISD readouts).

Outcome: an evidence-based method choice instead of a default (worked end-to-end in §2.5; real data in E4c).